INDIVIDUAL REFLECTION

Mini Use Case 02 – Credit Pre-Screening

Calibo AI Academy | Part 2 – KLU Batch | Phase 2 | Supervised Learning Part 1

Team Member	R. Mahima Varshini
Role	Data Analyst (Lead)
Contribution	CBIM Problem Canvas and Step 5.1 (A–D): Load, Inspect, Analyze, and Clean Data

1. Project Overview

Our team worked on Mini Use Case 02 in the BFSI – Retail Lending (Personal Loans) domain. The project focuses on understanding credit pre-screening data and preparing it for analysis and supervised machine learning. The dataset contained 12,000 loan applications and 19 columns covering demographics, employment, income, existing EMI, credit history, loan details, and historical decisions.

2. My Role and Contribution

I worked as the Data Analyst Lead. My assigned work included the CBIM Problem Canvas and Step 5.1 sections A–D. I loaded the dataset, inspected its structure, analyzed historical approval and decline outcomes, and investigated data-quality issues before the later analysis and modelling stages.

- **Data loading:** Imported the required libraries and loaded the CSV file using Pandas. I used `df.head()` to inspect the first five records.
- **Dataset overview:** Used `df.shape`, `df.columns`, and `df.info()` to check the number of rows and columns, column names, data types, and non-null counts.
- **Historical decisions:** Counted the approved and declined applications and calculated the overall approval and decline rates.
- **Data-quality checks:** Investigated missing monthly income, missing credit scores, and unusually high monthly-income values.
- **Cleaning documentation:** Documented the treatment of unusual income values, missing income, and missing credit scores, including the reason for each decision.

3. Key Findings

- The dataset contained 12,000 applications and 19 columns.
- There were 6,353 approved applications and 5,647 declined applications.
- The overall approval rate was 52.94%, and the decline rate was 47.06%.
- There were 441 missing monthly-income values. Missing income was concentrated among Business Owners and Self-Employed Professionals, so dropping these rows could disproportionately remove applicants from those groups.
- There were 1,039 missing credit scores. All these records had zero months of credit history, suggesting that they represented new-to-credit applicants in this dataset.

- Twelve monthly-income values were above ₹6,00,000, and all belonged to Salaried applicants. The notebook treated them as likely annual income entered into a monthly field and documented dividing those values by 12.
- The documented cleaning approach filled missing income using the median for the applicant's employment type, created an `is_ntc` indicator before filling missing credit scores with the overall median, and retained all 12,000 records.

4. What I Learned

- I learned that inspecting the dataset's structure and quality is necessary before conducting detailed analysis or training a model.
- I practised using Pandas functions such as `head()`, `shape`, `info()`, `isnull().sum()`, `value_counts()`, and `groupby()`.
- I learned to investigate the pattern and meaning of missing values before deciding how to handle them.
- I understood that a missing credit score may carry business meaning. In this dataset, it was associated with zero credit-history months, so the `is_ntc` indicator helped preserve that information.
- I gained a better understanding of data leakage and why model preprocessing statistics should be learned from training data only.

5. Challenges and How I Addressed Them

A key challenge was deciding how to handle missing values without unnecessarily losing useful records or creating misleading results. The missing income values were concentrated in particular employment groups, while missing credit scores were associated with zero credit-history months. I investigated these patterns and documented a reason for the cleaning decisions instead of simply deleting incomplete rows. I also reviewed unusually high income values and documented the likely annual-versus-monthly entry issue. These interpretations should be validated against a reliable source or business rule where available.

6. Teamwork and Business Understanding

My work established a clearer starting point for the team's later analysis and model-building tasks. By documenting the dataset structure, historical decision distribution, and data-quality findings, I helped make the analysis more transparent and reproducible. I also learned that historical approval and decline labels represent past analyst decisions; they do not by themselves prove an applicant's ability to repay a loan.

7. What I Would Improve Next Time

- Create a summary table showing missing-value counts and percentages before and after cleaning.
- Validate unusual income values against a documented business rule or reliable source before correcting them.
- Maintain a clear cleaning log recording each issue, the action taken, the reason, and the number of affected records.
- For machine learning, fit imputation steps on the training set only and apply the learned values to the test set to reduce data leakage.
- Consider fairness, reliability, and business risks before using a credit model to support real lending decisions.

8. Conclusion

This task strengthened my practical understanding of data inspection, exploratory analysis, data-quality assessment, and responsible preprocessing. My main learning is that successful machine learning starts with understanding the data and documenting decisions carefully, not just building a model. Through Step 5.1, I contributed the initial analysis and cleaning documentation for our Credit Pre-Screening use case.